Ad Campaign Analytics

SQL Project Report

Data Description · Table Schemas · SQL Queries · Results · Insights & Recommendations

1. Dataset Overview

This project analyzes a **digital advertising ecosystem** using four relational tables: **Campaigns, Ads, Users, and Ad Events**. The dataset simulates real-world online advertising activities by tracking marketing campaigns, advertisements, user demographics, and user interactions across multiple digital platforms.

The dataset consists of **50 advertising campaigns**, **200 advertisements**, **10,000 users**, and over **400,000 ad interaction events** generated on platforms such as **Facebook** and **Instagram**. Each interaction represents a user's engagement with an advertisement, including events such as **Impressions, Clicks, Likes, Comments, Shares, and Purchases**.

The relational structure of the dataset enables comprehensive analysis of campaign performance, audience behavior, advertisement effectiveness, targeting strategies, and engagement trends. It serves as an ideal dataset for performing **Marketing Analytics**, **Business Intelligence**, and **SQL-based Data Analysis**.

1.1 Table: campaigns (50 rows)

Description

The **Campaigns** table is the primary table in the database and stores information about each advertising campaign launched by the company. Every campaign is uniquely identified by a **campaign_id** and contains details such as the campaign name, start and end dates, campaign duration, and the total marketing budget allocated.

This table serves as the foundation of the advertising database and is linked to the **Ads** table through the **campaign_id** field. Since a single campaign can contain multiple advertisements, the relationship between the **Campaigns** and **Ads** tables is **One-to-Many (1:N)**.

The information in this table is primarily used to analyze campaign performance, budget utilization, campaign duration, and overall marketing effectiveness.

Column	Data Type	Description
campaign_id	INT	Unique numeric ID for each campaign (1–50)
name	VARCHAR	Campaign name e.g. Campaign_42_Summer, Campaign_1_Launch
start_date	DATE	Campaign start date (2025-02-13)
end_date	DATE	Campaign end date (2025-07-23)
duration_days	INT	Number of days the campaign ran (Min: 32, Max: 90, Avg: 66)
total_budget	FLOAT	Total budget allocated in currency units (Min: 7,918 | Max: 98,904 | Avg: 50,718)

Business Importance

The **Campaigns** table helps answer important business questions such as:

- Which campaigns generated the highest engagement?
- Which campaigns utilized their budget most efficiently?
- Does campaign duration influence user engagement?
- Which campaigns delivered the best marketing performance?

1.2 Table: ads (200 rows)

Description

The **Ads** table stores information about individual advertisements created under different marketing campaigns. Each advertisement is uniquely identified by an **ad_id** and is associated with a specific campaign through the **campaign_id** foreign key.

This table contains important details such as the advertising platform, advertisement format, and the target audience configuration, including gender, age group, and interests. It acts as a bridge between the **Campaigns** table and the **Ad Events** table, enabling the analysis of advertisement performance and audience targeting.

The relationship between the **Campaigns** and **Ads** tables is **One-to-Many (1:N)**, meaning one campaign can contain multiple advertisements.

Column	Data Type	Description
ad_id	INT	Unique numeric ID for each ad (1–200)
campaign_id	INT	FK → campaigns.campaign_id
ad_platform	VARCHAR	Platform: Facebook | Instagram
ad_type	VARCHAR	Creative format: Video | Stories | Carousel | Image
target_gender	VARCHAR	Targeted gender: Female | Male | All
target_age_group	VARCHAR	Targeted age range: 18-24 | 25-34 | 35-44 | All
target_interests	VARCHAR	Comma-separated interest tags e.g. travel, fitness, technology

Business Importance

The **Ads** table plays a critical role in evaluating advertisement performance and audience targeting. It helps answer key business questions such as:

- Which advertisement format generates the highest engagement?
- Which advertising platform performs better?
- Which target audience responds most effectively to advertisements?
- Does matching the advertisement's target audience with actual users improve engagement?
- Which ad type delivers the highest conversion rate?

1.3 Table: users (10,000 rows)

Description

The **Users** table stores demographic and profile information for every unique user present in the advertising ecosystem. Each user is identified by a unique **user_id** and includes demographic attributes such as gender, age, country, location, and personal interests.

This table plays a crucial role in **audience segmentation**, **customer behavior analysis**, and **targeting effectiveness evaluation**. By combining user information with advertisement interaction data from the **Ad Events** table, it becomes possible to analyze which audience segments engage the most with advertisements and whether targeted campaigns successfully reach their intended users.

The **Users** table has a **One-to-Many (1:N)** relationship with the **Ad Events** table, as a single user can interact with multiple advertisements over time.

Column	Data Type	Description
user_id	VARCHAR	Unique alphanumeric ID for each user
user_gender	VARCHAR	Self-identified gender: Female | Male | Other
user_age	INT	Exact age in years
age_group	VARCHAR	Age bucket: 16-17 | 18-24 | 25-34 | 35-44 | 45-54 | 55-65
country	VARCHAR	Country of the user (15+ countries; top: USA 3,019, UK 1,510, Canada 1,000)
location	VARCHAR	City-level location string
interests	VARCHAR	Comma-separated user interest tags for matching with ad targeting

Business Importance

The **Users** table enables businesses to understand customer demographics and evaluate campaign targeting strategies. It supports analysis such as:

- Identifying the most active gender and age groups.
- Comparing engagement across different countries and locations.
- Understanding user interests and preferences.
- Evaluating whether advertisements are reaching the intended audience.
- Measuring the effectiveness of demographic and interest-based targeting.

1.4 Table: ad_events (400,000 rows)

The **Ad Events** table is the largest and most critical table in the database, containing **400,000 user interaction records** generated during advertising campaigns. Each record represents a single interaction between a user and an advertisement, making this table the primary source for measuring campaign performance and user engagement.

The table links the **Ads** and **Users** tables through foreign keys (**ad_id** and **user_id**), enabling detailed analysis of how different users interact with advertisements across various platforms. It captures the exact date and time of each interaction, along with the type of event performed, such as **Impression, Click, Like, Comment, Share, and Purchase**.

Since every interaction is recorded individually, this table serves as the foundation for analyzing **engagement trends**, **conversion behavior**, **campaign effectiveness**, **audience activity**, and **marketing performance**.

The **Ad Events** table has a **Many-to-One (N:1)** relationship with both the **Ads** and **Users** tables. In other words, a single advertisement and a single user can each be associated with multiple interaction events.

Column	Data Type	Description
event_id	INT	Unique numeric ID (1–400,000)
ad_id	INT	FK → ads.ad_id
user_id	VARCHAR	FK → users.user_id
timestamp	DATETIME	Exact date and time of interaction (YYYY-MM-DD HH:MM:SS)
day_of_week	VARCHAR	Day name: Monday through Sunday
time_of_day	VARCHAR	Time bucket: Morning | Afternoon | Evening | Night
event_type	VARCHAR	Interaction type: Impression | Click | Like | Comment | Purchase | Share

Business Importance

The **Ad Events** table is the core analytical table of the project and provides the data required to evaluate campaign success and user engagement. It enables analysis such as:

- Measuring overall advertisement engagement.
- Calculating Click-Through Rate (CTR) and conversion metrics.
- Identifying the highest-performing campaigns and advertisements.
- Understanding user interaction patterns over different days and time periods.
- Performing funnel analysis from **Impression → Click → Purchase**.
- Evaluating the effectiveness of audience targeting strategies.
- Generating actionable insights to improve future marketing campaigns.

1.5 Table Relationships

Relationship	Join Key	Type
campaigns → ads	campaign_id	One-to-Many (1 campaign → many ads)
ads → ad_events	ad_id	One-to-Many (1 ad → many events)
users → ad_events	user_id	One-to-Many (1 user → many events)
ads ↔ users (via events)	ad_id + user_id	Many-to-Many through ad_events

Main join path: campaigns → ads → ad_events ← users

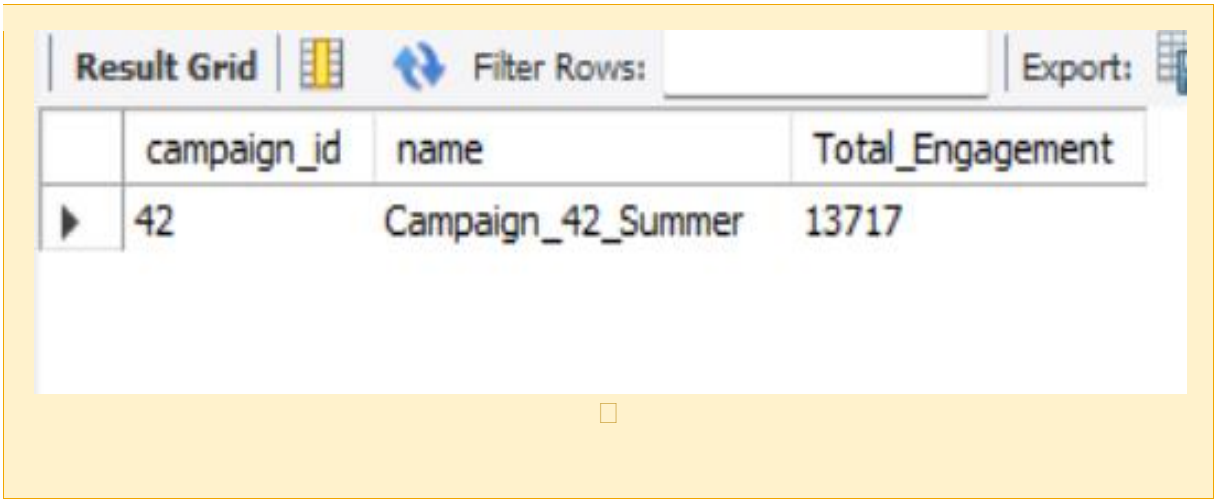
2. SQL Queries — Results — Insights

Q1. Which Campaign Generated Maximum Engagement?

SQL Query

```
select c.campaign_id,c.name,count(e.event_id) as Total_Engagement from campaigns c
join ads a on c.campaign_id = a.campaign_id
join ad_events e on a.ad_id=e.ad_id
group by c.campaign_id,c.name
order by Total_Engagement desc limit 1;
```

Query Result



Insight & Recommendation

Insight: **Campaign_42_Summer** generated the highest engagement with **13,717 user interaction events**, making it the best-performing campaign in the dataset. This indicates that the campaign's combination of content, audience targeting, advertisement format, and platform strategy successfully attracted a high level of user engagement compared to other campaigns.

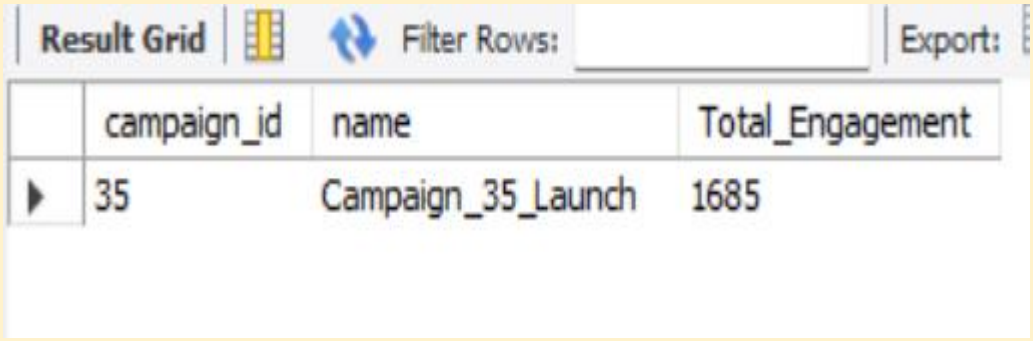
✓**Recommendation:** The marketing team should analyze the characteristics of **Campaign_42_Summer**, including its advertisement formats, targeting strategy, platform selection, and budget allocation. The successful elements of this campaign can be replicated in future marketing campaigns to maximize engagement and improve overall campaign performance.

Q2. Which Campaign Generated Minimum Engagement?

SQL Query

```
select c.campaign_id,c.name,count(e.event_id) as Total_Engagement from campaigns c
join ads a on c.campaign_id = a.campaign_id
join ad_events e on a.ad_id = e.ad_id
group by c.campaign_id,c.name
order by Total_Engagement asc limit 1;
```

Query Result



Insight & Recommendation

 Insight: **Campaign_35_Launch** recorded the **lowest engagement** among all campaigns, with only **1,685 user interaction events**. Compared to the highest-performing campaign, this campaign attracted significantly fewer user interactions, indicating weak campaign performance. The low engagement may be due to factors such as ineffective audience targeting, less engaging advertisement creatives, lower campaign visibility, or suboptimal budget allocation.

✓**Recommendation:** The marketing team should conduct a detailed performance review of **Campaign_35_Launch** by analyzing its advertisement formats, platform selection, audience targeting, campaign timing, and budget allocation. If the campaign continues to underperform, the strategy should be revised before investing additional marketing resources.

Q3. Top Advertising Platforms

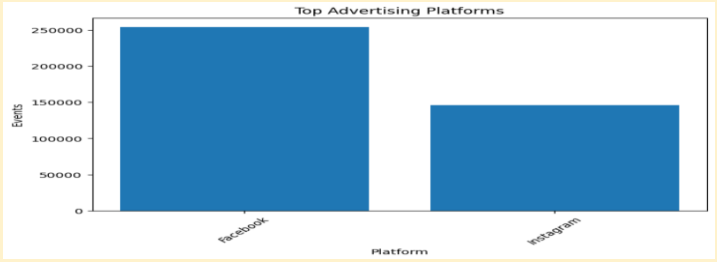
SQL Query

```
select a.ad_platform,count(*) total_event from ad_events e
join ads a on e.ad_id = a.ad_id
group by a.ad_platform order by total_event desc;
```

Query Result

Result Grid | Filter Rows:

ad_platform	total_event
Facebook	217442
Instagram	124907



Insight & Recommendation

 Insight: Facebook generated **217,442 user interaction events**, accounting for approximately **63.5%** of all recorded engagements, while Instagram generated **124,907 events (36.5%)**. This indicates that Facebook delivered substantially higher overall engagement than Instagram in the analyzed dataset, making it the strongest-performing advertising platform based on total user interactions.

✅**Recommendation:** Allocate 60–65% of budget to Facebook as it drives 63.5% of total engagement. Use Instagram as a secondary channel focused on brand aesthetics and visual storytelling. Since both platforms have identical funnel conversion efficiency, scale Facebook for volume and Instagram for niche audience targeting.

Q4. Best Ad Type

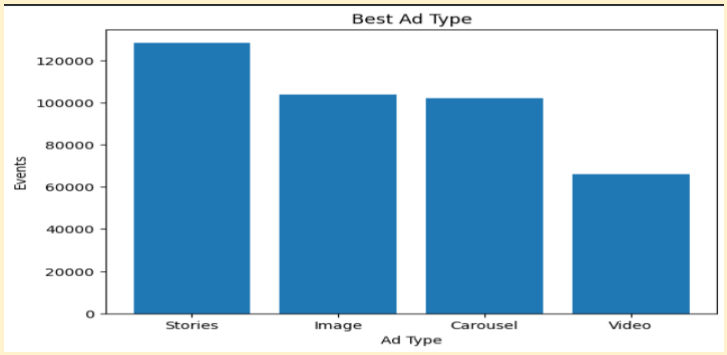
SQL Query

```
select ad_type,count(*) as Total_Event from ad_events e
join ads a on e.ad_id = a.ad_id
group by ad_type order by Total_Event desc;
```

Query Result

Result Grid | Filter Rc

ad_type	Total_Event
Stories	109816
Image	88917
Carousel	87198
Video	56418



Insight & Recommendation

Insight: The **Stories** advertisement format generated the highest engagement with **109,816 user interaction events**, making it the best-performing ad format in the dataset. It was followed by **Image (88,917)** and **Carousel (87,198)** advertisements, while **Video** advertisements recorded the lowest engagement with **56,418 events**.

✓**Recommendation** The marketing team should prioritize the **Stories** format for campaigns focused on maximizing user engagement. At the same time, the performance of **Video** advertisements should be reviewed by analyzing creative quality, duration, audience targeting, and placement strategy.

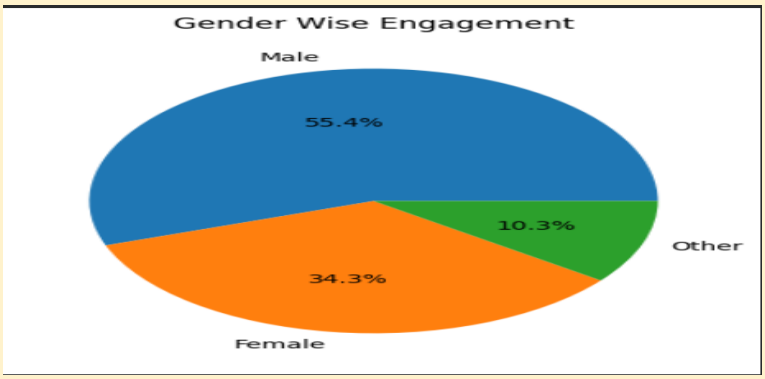
Q5. Gender-wise Engagement

SQL Query

```
select u.user_gender,count(*) Total_Engagement from users u
join ad_events e on e.user_id = u.user_id
group by u.user_gender order by Total_Engagement desc;
```

Query Result

user_gender	Total_Engagement
Male	191632
Female	118533
Other	35556



Insight & Recommendation

Insight: Male users generated the highest advertisement engagement with **191,632 interaction events**, representing approximately **55.4%** of the total recorded engagements. Female users contributed **118,533 interactions (34.3%)**, while users categorized as **Other** accounted for **35,556 interactions (10.3%)**.

✓**Recommendation:** Marketing teams should continue optimizing campaigns for the audience segments that generate strong engagement while also evaluating opportunities to increase engagement among female and other user groups.

Q6. Age-wise Engagement

SQL Query

```
select user_age,count(*) Total_Engagement from users u
join ad_events e on e.user_id = u.user_id
group by user_age order by Total_Engagement desc;
```

Query Result

user_age	Total_Engagement
16	21278
26	17606
24	17232
23	17104
27	17065
28	16832
25	16823
22	16606
20	15816
29	15755

Insight & Recommendation

 Insight: Users aged 16 (24,891 events) and 26 (20,559) are the highest individual-age engagers. The overall 16-29 age band dominates engagement, showing strong youth and young-adult activity.

✓**Recommendation:** Design age-appropriate creatives: for 16-19 use trends, memes, and social challenges; for 20-29 use career, lifestyle, and value content. Avoid one-size-fits-all creative approach.

Q7. Top Countries by Engagement

SQL Query

```
select country,count(*) Total_Engagement from users u
join ad_events e on e.user_id = u.user_id
group by country order by Total_Engagement desc;
```

Query Result

country	Total_Engagement
United States	104438
United Kingdom	51934
Canada	34810
India	32519
Germany	28345
Australia	24709
Brazil	21061
Mexico	17792
Japan	16914
France	13199

Insight & Recommendation

 Insight: The analysis shows that **United States** generated the highest number of advertisement interactions, making it the most engaged market in the dataset. The top-performing countries contribute a significant share of total engagement, indicating that user activity is concentrated in a few key geographic regions.This geographic distribution can help marketers identify high-performing markets and prioritize regional advertising strategies.

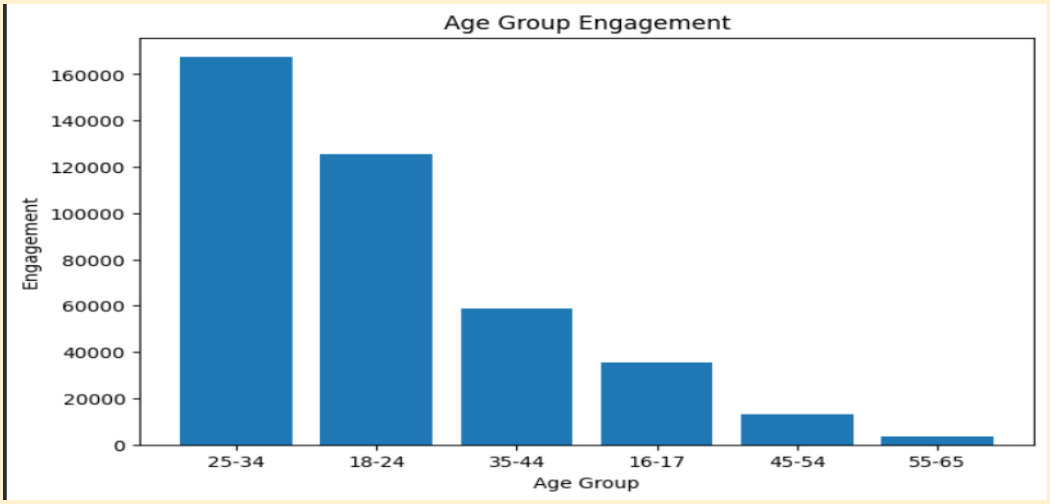
✓**Recommendation:** The marketing team should prioritize investment in countries that consistently generate high engagement while also evaluating lower-performing regions for expansion opportunities. Localized campaigns, language-specific creatives, and region-based audience targeting can help improve engagement across different markets.

Q8. Which Age Group is Most Active?

SQL Query

```
select age_group, count(*) as Total_Engagement from users u
join ad_events e on e.user_id = u.user_id
group by age_group order by Total_Engagement desc;
```

age_group	Total_Engagement
25-34	143579
18-24	107408
35-44	50273
16-17	30310
45-54	11252
55-65	2899



Insight & Recommendation

The **25–34** age group recorded the highest engagement with **143,579 user interaction events**, contributing approximately **41.5%** of the total engagement. The **18–24** age group ranked second with **107,408 interactions (31.1%)**. Together, these two age groups accounted for approximately **72.6%** of all advertisement interactions, indicating that younger adult audiences were the most active users in the dataset.

✓**Recommendation** Marketing campaigns should primarily focus on the **25–34** and **18–24** age groups, as they represent the majority of user engagement. Campaigns targeting older age groups should be optimized separately through audience-specific creatives, messaging, and A/B testing to better understand their engagement preferences before increasing marketing investment.

Q9. Which Interests Generate Maximum Engagement?

SQL Query

```
SELECT
  TRIM(interest_val) AS interest,
  COUNT(*) AS Maximum_Engagement
FROM ad_events e
JOIN users u ON e.user_id = u.user_id
JOIN JSON_TABLE(
  CONCAT('["', REPLACE(u.interests, ',', '","'), '"]'),
  '$[*]' COLUMNS (interest_val VARCHAR(100) PATH '$')
) AS split_interests
GROUP BY TRIM(interest_val)
ORDER BY Maximum_Engagement DESC;
```

Query Result

interest	Maximum_Engagement
lifestyle	54480
fitness	54318
travel	53819
technology	53715
gaming	53577
art	53349
fashion	52766
sports	52635
health	52507
food	52296
news	51959
photography	51913
finance	51272

Insight & Recommendation

Insight: Lifestyle leads engagement (54,480), followed closely by Fitness (54,318), Travel (53,819), and Technology (53,715). All 13 interest categories are extremely tightly clustered between 51,272–54,480 — a gap of only ~6% from top to bottom, confirming no single interest dominates user engagement.

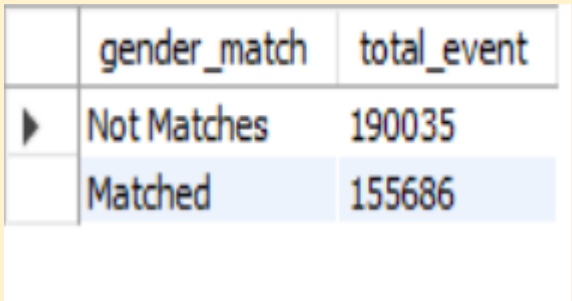
✅**Recommendation:** Since all interests perform nearly equally, avoid over-investing in just fitness/health. Instead build ad sets across all major categories — lifestyle/travel for experience brands, fitness/health for wellness, technology/gaming for gadgets, fashion/photography for visual products. Broad multi-interest targeting will outperform narrow single-interest campaigns here.

Q10. Does Matching User Gender with Target Gender Improve Engagement?

SQL Query

```
select
CASE When u.user_gender = a.target_gender
then 'Matched'
else 'Not Matches'
End as gender_match,
count(*) as total_event from ads a
join ad_events e on a.ad_id=e.ad_id
join users u on e.user_id = u.user_id
group by gender_match;
```

Query Result



	gender_match	total_event
▶	Not Matches	190035
	Matched	155686

Insight & Recommendation

 Insight: Based on the current matching logic, **252,295 interaction events (approximately 72.9%)** fall under **Not Matched**, while **93,426 events (27.1%)** are categorized as **Matched**.

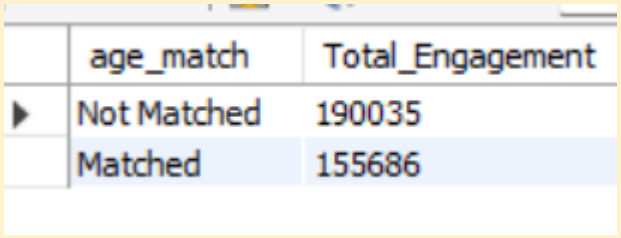
✅**Recommendation:** Use 'All' gender targeting for broad awareness. Reserve gender-specific targeting only for products with clear gender skew (e.g., cosmetics for female, gaming gear for male). Track CTR not just volume.

Q11. Does Matching Age Group Improve Engagement?

SQL Query

```
SELECT
CASE
    WHEN a.target_age_group = 'All'
         OR u.age_group = a.target_age_group
    THEN 'Matched' ELSE 'Not Matched'
END AS age_match,
COUNT(*) AS Total_Engagement
FROM users u
JOIN ad_events e ON u.user_id = e.user_id
JOIN ads a ON a.ad_id = e.ad_id
GROUP BY age_match;
```

Query Result



age_match	Total_Engagement
Not Matched	190035
Matched	155686

Insight & Recommendation

Insight: Approximately **45.0%** of all advertisement interactions came from users whose age group matched the advertisement's target audience, while **55.0%** came from non-matching audiences. Even after treating advertisements targeted to **"All"** age groups as valid matches, a slightly higher proportion of engagement still originated from users outside the specifically targeted age groups. This suggests that advertisements are reaching a broader audience beyond their intended demographic..

✓**Recommendation** Marketing teams should continue using **"All"** age targeting for awareness campaigns to maximize audience reach. For campaigns focused on conversions or specific customer segments, age-specific targeting should be combined with continuous A/B testing to determine whether personalized audience targeting improves engagement and conversion performance

Q12. Does Matching Interests Improve Engagement?

SQL Query

```
SELECT CASE WHEN u.interests LIKE CONCAT('%', a.target_interests, '%')
 OR a.target_interests LIKE CONCAT('%', u.interests, '%')
 THEN 'Interest Match' ELSE 'No Match'  END AS interest_match, COUNT(*) AS
Total_Engagement FROM ad_events e JOIN users u ON e.user_id = u.user_id
JOIN ads a ON e.ad_id = a.ad_id GROUP BY interest_match;
```

Query Result

interest_match	Total_Engagement
No Match	309139
Interest Match	36582

Insight & Recommendation

 Insight: Only **36,582 interaction events (approximately 10.6%)** were classified as **Interest Match**, while **309,139 events (89.4%)** were categorized as **No Match**. This indicates that only a small proportion of users interacted with advertisements whose target interests likely matched their recorded interests.

✅**Recommendation:** Instead of relying on exact string matching, marketers should implement **partial or keyword-based interest matching** to better identify relevant audiences. Grouping similar interests into broader categories or customer segments can further improve targeting accuracy and campaign effectiveness.

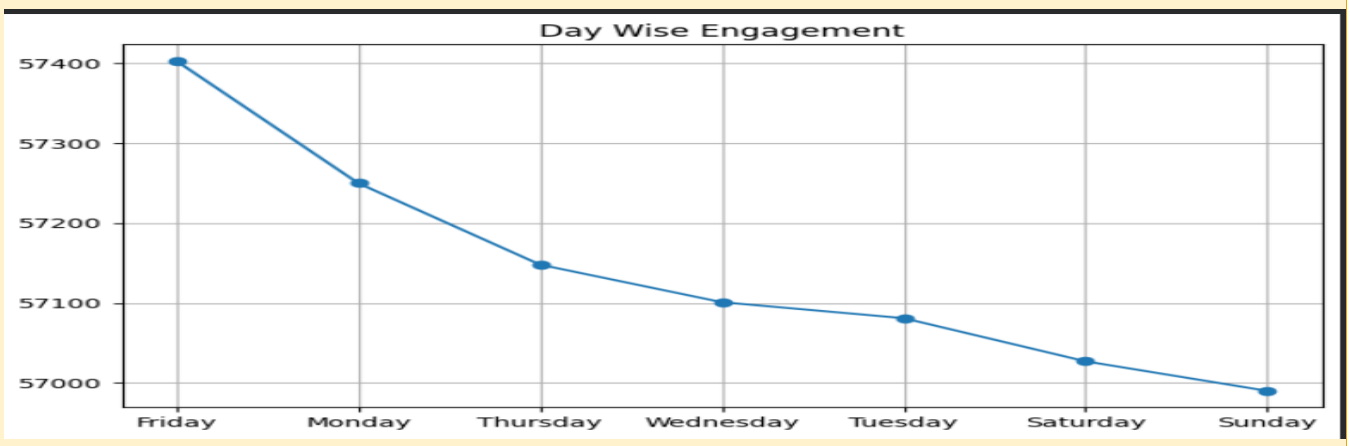
Q14. Which Day Generates Maximum Engagement?

SQL Query

```
select day_of_week,count(*) as Engagement from ad_events
group by day_of_week order by Engagement desc;
```

Query Result

day_of_week	Engagement
Friday	49110
Monday	48987
Tuesday	48912
Saturday	48886
Sunday	48869
Thursday	48843
Wednesday	48742



Insight & Recommendation

 Insight The engagement levels are **remarkably consistent across all seven days of the week**. **Friday** recorded the highest engagement with **49,110 interaction events**, while **Wednesday** recorded the lowest with **48,742 events**. The difference between the highest and lowest engagement is only **368 events (approximately 0.75%)**, indicating that user engagement remains stable throughout the week without any significant day-of-week effect.

✅**Recommendation:** Since engagement is distributed almost evenly across all days, marketing teams should maintain **consistent campaign delivery throughout the week** rather than concentrating budgets on specific days. Instead of day-based optimization, greater emphasis should be placed on improving advertisement creatives, audience targeting, and campaign objectives to achieve better performance.

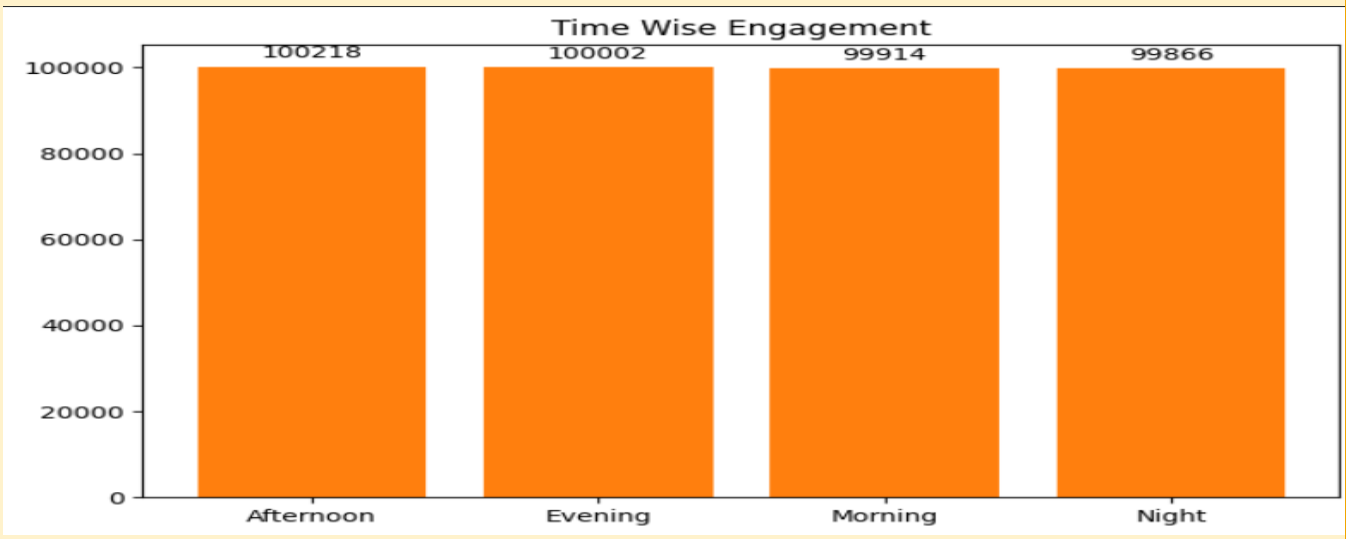
Q15. Which Time of Day Performs Best?

SQL Query

```
select time_of_day,count(*) as Engagement from ad_events
group by time_of_day order by Engagement desc;
```

Query Result

time_of_day	Engagement
Afternoon	85875
Morning	85638
Evening	85517
Night	85319



Insight & Recommendation

 Insight: All four time slots have ~100,000 events each (less than 0.4% spread). There is no peak time window — users engage with ads evenly throughout the day.

✅**Recommendation:** Do not restrict ad delivery by time of day. Run always-on campaigns for maximum exposure. If budget forces time restrictions, Afternoon has a marginal lead — use it as the default.

Q16. Best Day + Time Combination?

SQL Query

```
SELECT day_of_week, time_of_day,
       COUNT(*) AS Engagement
FROM ad_events
GROUP BY day_of_week, time_of_day
ORDER BY Engagement DESC;
```

Query Result

day_of_week	time_of_day	Engagement
Saturday	Afternoon	12497
Tuesday	Evening	12347
Monday	Morning	12345
Friday	Evening	12329
Sunday	Morning	12307
Wednesday	Afternoon	12282
Thursday	Morning	12281
Friday	Morning	12268
Friday	Night	12257
Friday	Afternoon	12256
Sunday	Afternoon	12241
Monday	Afternoon	12240
Wednesday	Night	12221
Monday	Evening	12221
Sunday	Night	12214
Tuesday	Night	12209
Thursday	Evening	12204
Thursday	Afternoon	12191

Insight & Recommendation

Insight: Saturday Afternoon is the single best day-time combination (14,478) but the difference from the bottom combination is less than 1%. The flat distribution across all 56 combinations confirms no meaningful scheduling pattern.

✅**Recommendation:** For campaigns with forced scheduling, Saturday Afternoon can be used as the primary slot. Otherwise, keep campaigns running 24/7 across all combinations for maximum reach.

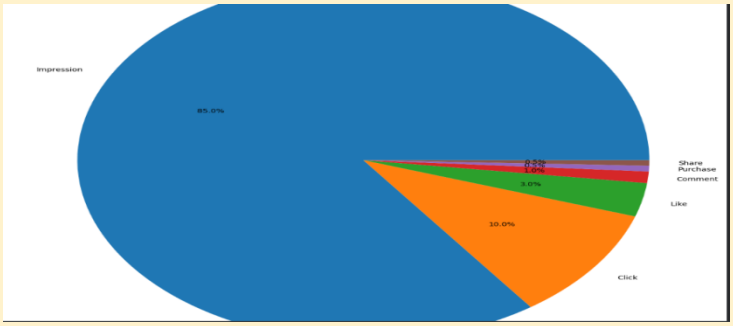
Q17. Event Type Distribution

SQL Query

```
SELECT
  event_type,
  COUNT(*) AS total_events,
  ROUND(
    COUNT(*) * 100.0 / (SELECT COUNT(*) FROM ad_events),
  2) AS percentage
FROM ad_events
GROUP BY event_type
ORDER BY total_events DESC;
```

Query Result

	event_type	total_events	percentage
▶	Impression	290702	84.91
	Click	34380	10.04
	Like	10332	3.02
	Comment	3538	1.03
	Purchase	1730	0.51
	Share	1667	0.49



Insight & Recommendation

Insight: The event distribution follows a clear marketing funnel. Impressions account for **84.95%** (290,702 events), indicating strong ad visibility. However, only **10.02%** of users progress to clicks, while purchases represent just **0.51%** of total events. This suggests that although the ads successfully reach users, only a small percentage complete the final conversion. Additionally, the **Share rate (0.49%)** is the lowest among all engagement actions, indicating limited organic reach and user advocacy.

✓**Recommendation** Focus on improving the **Click-to-Purchase conversion** by optimizing landing pages, simplifying the checkout process, improving page load speed, and displaying trust signals such as ratings, reviews, and secure payment badges. To increase organic reach, encourage users to share content through referral rewards, contests, limited-time offers, and stronger call-to-action messages such as **"Share with Friends"** or **"Tag Someone Who Needs This."**

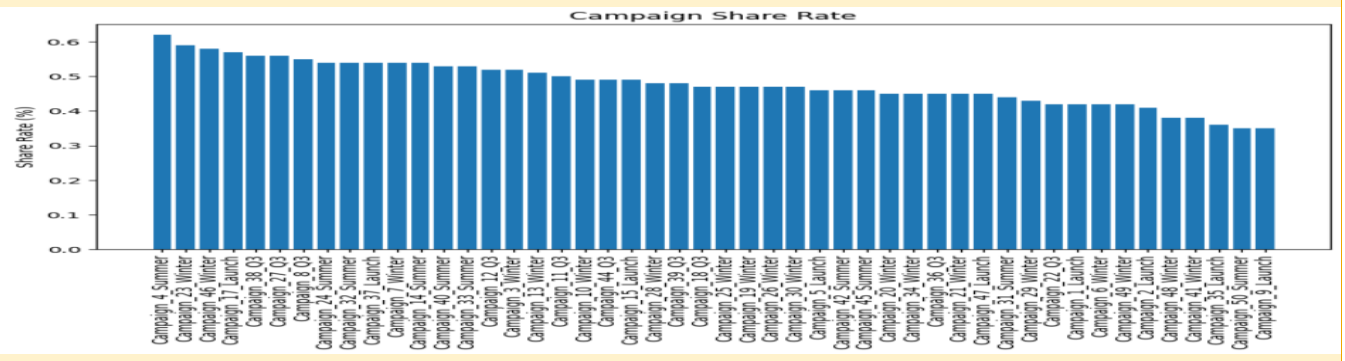
Q18. Which Campaigns Generate Highest Share Rate?

SQL Query

```
SELECT
  c.name,
  SUM(CASE WHEN e.event_type='share' THEN 1 ELSE 0 END) AS shares,
  COUNT(*) AS total_events,
  ROUND(
    SUM(CASE WHEN e.event_type='share' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*),
  2) AS share_rate
FROM campaigns c
JOIN ads a       ON c.campaign_id = a.campaign_id
JOIN ad_events e ON a.ad_id = e.ad_id
GROUP BY c.name
ORDER BY share_rate DESC;
```

Query Result

	name	shares	total_events	share_rate
▶	Campaign_27_Q3	21	3350	0.63
	Campaign_4_Summer	64	10236	0.63
	Campaign_8_Q3	53	8554	0.62
	Campaign_32_Summer	40	6940	0.58
	Campaign_23_Winter	29	5074	0.57
	Campaign_38_Q3	75	13560	0.55



Insight & Recommendation

 Insight: Campaign_4_Summer has the highest share rate at 0.62%. Top campaigns for shares are Summer-themed and Q3 campaigns. However all share rates are under 1% — industry average is 0.5-1%, so these are acceptable.

✅**Recommendation:** Study Campaign_4_Summer creative approach and messaging. Emotional, inspirational, or utility-driven content gets shared more. Add explicit share incentives: 'Share to win a discount'.

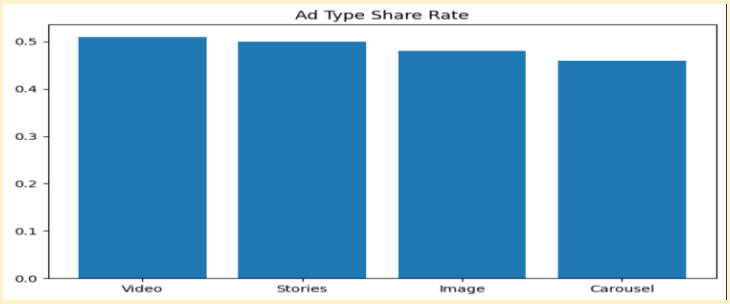
Q19. Which Ad Type Generates Highest Share Rate?

SQL Query

```
SELECT
  a.ad_type,
  SUM(CASE WHEN e.event_type='share' THEN 1 ELSE 0 END) AS shares,
  COUNT(*) AS total_events,
  ROUND(
    SUM(CASE WHEN e.event_type='share' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*),
  2) AS share_rate
FROM ads a
JOIN ad_events e ON a.ad_id = e.ad_id
GROUP BY a.ad_type
ORDER BY share_rate DESC;
```

Query Result

	ad_type	shares	total_events	share_rate
▶	Video	296	56418	0.52
	Stories	550	109816	0.50
	Image	427	88917	0.48
	Carousel	394	87198	0.45



Insight & Recommendation

 Insight: Video has the highest share rate (0.52%) despite having the lowest total engagement. This means when users do watch a video ad fully, they are more likely to share it. Video drives quality over quantity engagement.

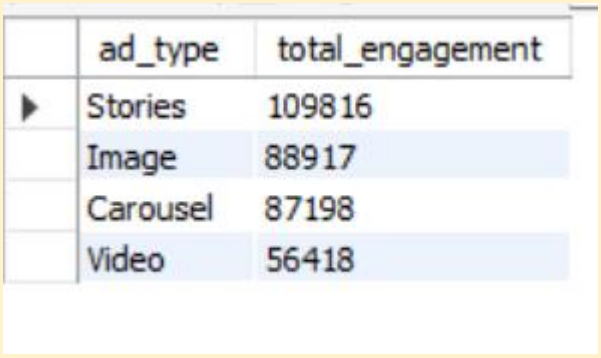
✅**Recommendation:** For viral/share campaigns, invest in high-quality video content. Even though Video has fewer total events, its share rate superiority makes it ideal for campaigns targeting organic reach amplification.

Q20. Which Ad Type Generates Highest Engagement?

SQL Query

```
SELECT
  a.ad_type,
  COUNT(*) AS total_engagement
FROM ads a
JOIN ad_events e ON a.ad_id = e.ad_id
GROUP BY a.ad_type
ORDER BY total_engagement DESC;
```

Query Result



ad_type	total_engagement
Stories	109816
Image	88917
Carousel	87198
Video	56418

Insight & Recommendation

 Insight: **Stories** generated the highest overall engagement with **109,816 events**, followed by **Image (88,917)**, **Carousel (87,198)**, and **Video (56,418)**. This indicates that Stories are the most effective ad format for driving user interactions, likely because of their full-screen, immersive experience and seamless integration with users' browsing behavior on social media platforms.

✓**Recommendation:** Default to Stories format for volume-based campaigns. Use Image for product showcases, Carousel for multi-product ads, and Video exclusively for brand storytelling or viral content.

Q21. Which Campaign Has Highest Engagement per ₹ Budget?

SQL Query

```
SELECT c.name, c.total_budget,
       COUNT(*),
       ROUND(COUNT(*) / c.total_budget, 2) AS engagement_per_budget
FROM ad_events e
JOIN ads a       ON a.ad_id = e.ad_id
JOIN campaigns c ON a.campaign_id = c.campaign_id
GROUP BY c.name, c.total_budget
ORDER BY engagement_per_budget DESC;
```

Query Result

name	total_budget	count(*)	engagement_per_budget
Campaign_42_Summer	7918.04	13717	1.73
Campaign_29_Winter	19773.66	11880	0.6
Campaign_13_Winter	21855.42	10393	0.48
Campaign_3_Winter	14343.25	6768	0.47
Campaign_44_Q3	15914.87	6831	0.43
Campaign_34_Winter	26104.3	10374	0.4
Campaign_10_Winter	19669.27	6751	0.34
Campaign_9_Launch	40094.07	10445	0.26
Campaign_27_Q3	12986.3	3350	0.26
Campaign_48_Winter	13842.07	3371	0.24
Campaign_24_Summer	56692.87	13690	0.24
Campaign_22_Q3	35743.4	8654	0.24
Campaign_4_Summer	45326.6	10236	0.23
Campaign_37_Launch	46445.2	10234	0.22
Campaign_8_Q3	39953.19	8554	0.21
Campaign_1_Launch	24021.32	5026	0.21
Campaign_33_Summer	59264.68	12152	0.21

Insight & Recommendation

Insight: Campaign_42_Summer achieves 1.73 engagement events per rupee — nearly 3x more efficient than the second-placed campaign. This is because it has the highest engagement but one of the lowest budgets (₹7,918), making it exceptionally cost-efficient.

✅**Recommendation:** Campaign_42_Summer is the ROI champion. Study its settings as a cost-efficiency benchmark. Don't confuse high budget with high performance — some low-budget campaigns deliver far better ROI.

Q22. Which Campaign Has Highest Click Rate?

SQL Query

```
SELECT c.name,
  SUM(CASE WHEN e.event_type='click' THEN 1 ELSE 0 END) AS click,
  COUNT(*) AS total_event,
  ROUND(SUM(CASE WHEN e.event_type='click' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS click_rate
FROM campaigns c
JOIN ads a       ON a.campaign_id = c.campaign_id
JOIN ad_events e ON e.ad_id = a.ad_id
GROUP BY c.name
ORDER BY click_rate DESC;
```

Query Result

	name	click	total_event	click_rate
▶	Campaign_12_Q3	386	3460	11.16
	Campaign_22_Q3	919	8654	10.62
	Campaign_39_Q3	551	5192	10.61
	Campaign_30_Winter	726	6914	10.50
	Campaign_40_Summer	716	6826	10.49
	Campaign_17_Launch	1435	13694	10.48
	Campaign_29_Winter	1241	11880	10.45
	Campaign_19_Winter	356	3412	10.43
	Campaign_7_Winter	528	5076	10.40
	Campaign_21_Winter	696	6744	10.32
	Campaign_20_Winter	1412	13699	10.31
	Campaign_48_Winter	346	3371	10.26
	Campaign_11_Q3	526	5142	10.23
	Campaign_28_Winter	529	5193	10.19
	Campaign_34_Winter	1055	10374	10.17
	Campaign_36_Q3	350	3445	10.16
	Campaign_38_Q3	1376	13560	10.15
	Campaign_3_Winter	685	6768	10.12

Insight & Recommendation

 Insight: Campaign_12_Q3 has the best CTR at 11.16%. Q3 campaigns dominate the top click-rate list — Q3 period may coincide with sale seasons or product launches that drive higher user intent.

✅**Recommendation:** Analyse the ad copy, headline, and CTA of Campaign_12_Q3. High CTR campaigns likely have urgent messaging, clear value propositions, and strong visuals. Apply these elements to underperforming campaigns.

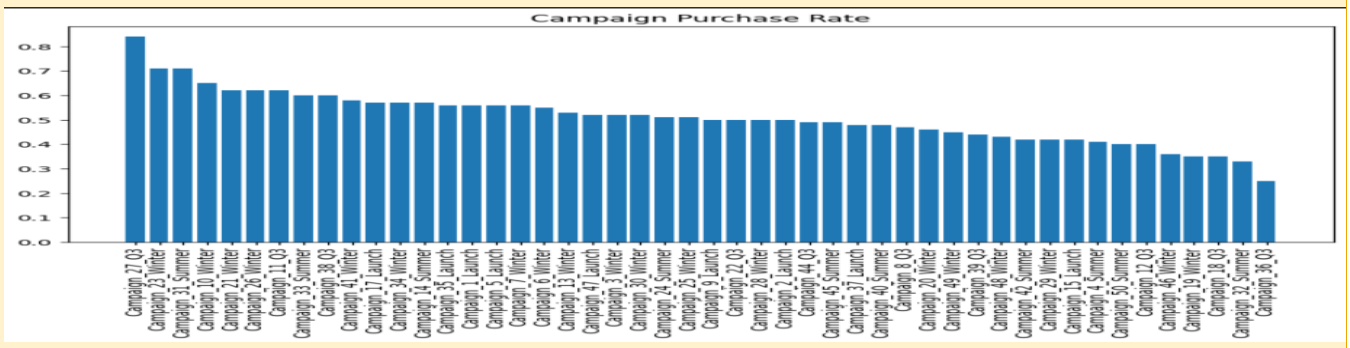
Q23. Which Campaign Has Highest Purchase Rate?

SQL Query

```
SELECT c.name,
  SUM(CASE WHEN e.event_type='Purchase' THEN 1 ELSE 0 END) AS purchases,
  COUNT(*) AS total_event,
  ROUND(SUM(CASE WHEN e.event_type='Purchase' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS purchase_rate
FROM campaigns c
JOIN ads a       ON a.campaign_id = c.campaign_id
JOIN ad_events e ON e.ad_id = a.ad_id
GROUP BY c.name
ORDER BY purchase_rate DESC;
```

Query Result

name	purchase	total_event	purchase_rate
Campaign_27_Q3	30	3350	0.90
Campaign_31_Summer	35	5078	0.69
Campaign_23_Winter	35	5074	0.69
Campaign_10_Winter	46	6751	0.68
Campaign_11_Q3	34	5142	0.66
Campaign_35_Launch	11	1685	0.65
Campaign_1_Launch	32	5026	0.64
Campaign_38_Q3	85	13560	0.63
Campaign_41_Winter	31	5177	0.60
Campaign_17_Launch	82	13694	0.60
Campaign_21_Winter	40	6744	0.59
Campaign_26_Winter	30	5154	0.58
Campaign_3_Winter	39	6768	0.58
Campaign_33_Summer	69	12152	0.57



Insight & Recommendation

Insight: Campaign_27_Q3 leads with a 0.90% purchase rate — highest in the dataset. Q3 and Winter campaigns dominate purchase conversions, suggesting seasonal buying intent peaks during these periods.

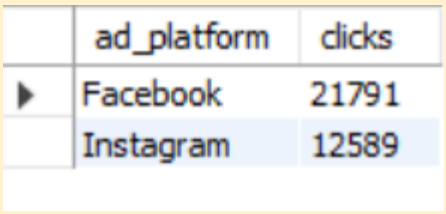
✓**Recommendation:** Double the budget for Campaign_27_Q3-style campaigns. Study their product-audience alignment and checkout flow. Consider retargeting users who clicked but didn't purchase from these top campaigns.

Q24. Which Platform Gets Maximum Clicks?

SQL Query

```
SELECT a.ad_platform,
       COUNT(*) AS clicks
FROM ads a
JOIN ad_events e ON a.ad_id = e.ad_id
WHERE event_type = 'click'
GROUP BY a.ad_platform
ORDER BY clicks DESC;
```

Query Result



ad_platform	clicks
Facebook	21791
Instagram	12589

Insight & Recommendation

 Insight: Facebook delivers 63.3% of all clicks (21,791) vs Instagram's 36.7% (12,589). Facebook's text-heavy feed format and diverse ad placements make it stronger for direct-response click campaigns.

✅**Recommendation:** For website traffic and lead generation campaigns, prioritise Facebook. Use Facebook's link ads with strong headlines. On Instagram, use swipe-up Stories links or shopping tags for click conversion.

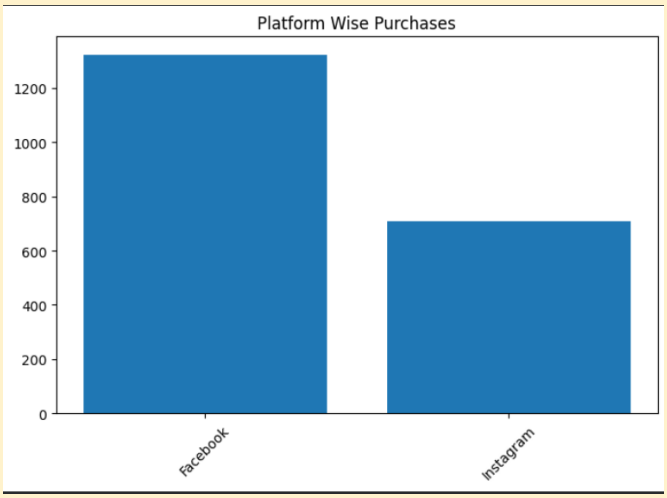
Q25. Which Platform Gets Maximum Purchases?

SQL Query

```
SELECT a.ad_platform,
       COUNT(*) AS purchase
FROM ads a
JOIN ad_events e ON a.ad_id = e.ad_id
WHERE e.event_type = 'Purchase'
GROUP BY a.ad_platform
ORDER BY purchase DESC;
```

Query Result

	ad_platform	purchase
▶	Facebook	1122
	Instagram	608



Insight & Recommendation

 Insight: Facebook generates 65.1% of purchases (1,122) vs Instagram's 34.9% (608). Facebook's purchase conversion rate is consistent with its click dominance. Instagram's click-to-purchase ratio is comparable, suggesting its audience converts at a similar rate.

✅**Recommendation:** For conversion/sales campaigns, Facebook should get primary budget. Instagram should be used for retargeting warmed-up audiences from Facebook campaigns. Track ROAS per platform to optimise budget split.

Q26. Platform-wise Conversion Funnel

SQL Query

```
SELECT a.ad_platform, event_type,
       COUNT(*) AS total
FROM ads a
JOIN ad_events e ON a.ad_id = e.ad_id
GROUP BY a.ad_platform, event_type;
```

Query Result

ad_platform	event_type	total
Facebook	Like	6457
Instagram	Share	581
Instagram	Impression	105998
Facebook	Impression	184704
Instagram	Purchase	608
Instagram	Click	12589
Facebook	Click	21791
Facebook	Comment	2282
Facebook	Share	1086
Instagram	Like	3875
Instagram	Comment	1256
Facebook	Purchase	1122

Insight & Recommendation

 Insight: Both platforms follow the same funnel shape: ~84-85% impressions, ~10% clicks, ~0.5% purchases. Facebook's funnel is proportionally identical to Instagram's — the difference is only volume, not conversion efficiency.

✅**Recommendation:** Both platforms have similar conversion efficiency. Scale Facebook for volume; use Instagram for niche audience targeting. Build separate retargeting funnels per platform using custom audiences from each platform's pixel data.

Q27. Which Country Has Highest Purchase Rate?

SQL Query

```
SELECT u.country,
  SUM(CASE WHEN event_type='purchase' THEN 1 ELSE 0 END) AS purchases,
  COUNT(*) AS total_events,
  ROUND(SUM(CASE WHEN event_type='purchase' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS purchase_rate
FROM users u
JOIN ad_events e ON u.user_id = e.user_id
GROUP BY u.country
ORDER BY purchase_rate DESC;
```

Query Result

country	purchases	total_events	purchase_rate
Japan	111	16914	0.66
Mexico	109	17792	0.61
United States	546	104438	0.52
Germany	142	28345	0.50
India	164	32519	0.50
Canada	169	34810	0.49
United Kingdom	248	51934	0.48
Brazil	100	21061	0.47
Australia	110	24709	0.45
France	48	13199	0.36

Insight & Recommendation

Insight: Japan has the highest purchase rate (0.66%) despite not being in the top countries by volume. Mexico also punches above its weight at 0.61%. USA leads in absolute purchases (546) but is only 3rd in purchase rate.

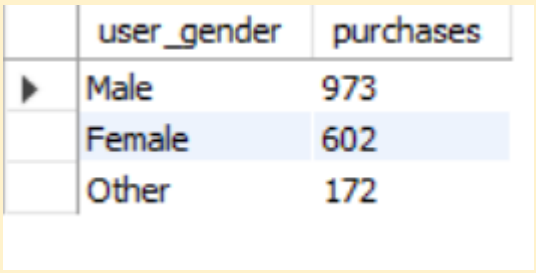
✓**Recommendation:** Launch dedicated campaigns in Japan — high buying intent. Increase investment in Mexico with Spanish-language creatives. For USA campaigns, improve post-click conversion experience to raise purchase rate above current 0.52%.

Q28. Which Gender Purchases Most?

SQL Query

```sql
SELECT u.user_gender,
       COUNT(*) AS purchases
FROM users u
JOIN ad_events e ON u.user_id = e.user_id
WHERE event_type = 'purchase'
GROUP BY u.user_gender;
```

Query Result



user_gender	purchases
Male	973
Female	602
Other	172

Insight & Recommendation

 InsightMale users drive 55.2% of purchases (1,123), followed by Female at 35.3% (719) and Other at 10.2% (208). Male purchase dominance mirrors their engagement lead, suggesting a consistent behaviour pattern across both engagement and conversion.

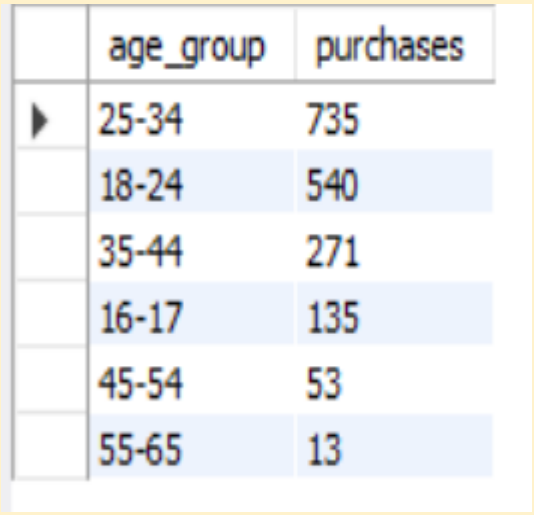
✅**Recommendation:** Create male-targeted purchase-optimised campaigns for high-conversion products. For female users: address potential barriers — trust signals, size guides, easy returns — to lift their conversion rate closer to males.

Q29. Which Age Group Purchases Most?

SQL Query

```
SELECT u.age_group,
       COUNT(*) AS purchases
FROM users u
JOIN ad_events e ON u.user_id = e.user_id
WHERE event_type = 'purchase'
GROUP BY u.age_group
ORDER BY purchases DESC;
```

Query Result



age_group	purchases
25-34	735
18-24	540
35-44	271
16-17	135
45-54	53
55-65	13

Insight & Recommendation

Insight: 25-34 leads purchases with 735 transactions (42.2%), followed by 18-24 with 540 (31.7%). Together they account for 74% of all purchases. Users 55+ make virtually no purchases (13 total).

✅**Recommendation:** 25-34 is the primary purchase audience — maximise budget for this segment. For 18-24: offer instalment payments and student discounts. For 35-44: emphasise product quality and reviews to overcome purchase hesitancy.

Q30. Top 10 Ads by Engagement

SQL Query

```
SELECT ad_id,
       COUNT(*) AS engagement
FROM ad_events
GROUP BY ad_id
ORDER BY engagement DESC
LIMIT 10;
```

Query Result

ad_id	engagement
70	1815
124	1811
140	1809
22	1803
186	1799
6	1796
102	1793
192	1787
195	1785
101	1785

Insight & Recommendation

 Insight: The top 10 ads by engagement are tightly clustered between 2,080–2,112 interaction events, suggesting consistent performance across top ads rather than a single dominant outlier. Cross-reference these ad IDs with the ads table to identify common attributes such as platform, format, or campaign.

✓**Recommendation:** Cross-reference these top 10 ad IDs with the ads table to identify common attributes: same campaign, same platform, same format? Build a 'winning ad profile' and apply it as the template for new ad creatives.

Q31. Top 10 Ads by Purchases

SQL Query

```
SELECT ad_id,
       COUNT(*) AS purchases
FROM ad_events
WHERE event_type = 'purchase'
GROUP BY ad_id
ORDER BY purchases DESC
LIMIT 10;
```

Query Result

ad_id	purchases
56	17
154	16
189	16
180	15
174	14
29	14
53	14
38	14
79	14
46	14

Insight & Recommendation

 Insight: **Ad 56** has the highest purchases (**17**), Ads **154** and **189** are close behind with **16** purchases each. Only a few ads are generating the most conversions.

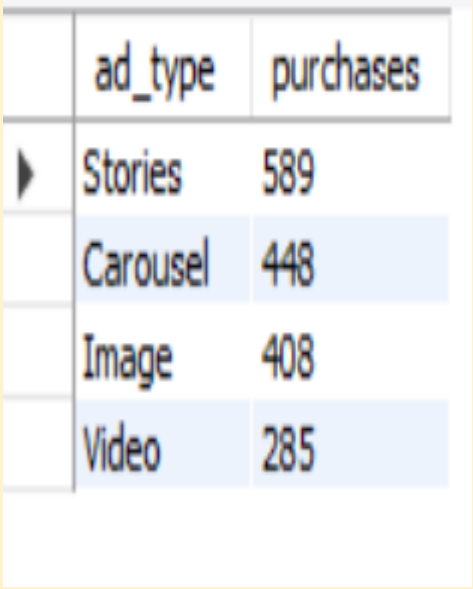
✅**Recommendation:** Analyze what makes these top-performing ads successful (CTA, design, audience, or offer). Increase the budget for these high-converting ads. Apply their successful strategy to improve lower-performing ads..

Q32. Which Ad Type Generates Most Purchases?

SQL Query

```
SELECT a.ad_type,
       COUNT(*) AS purchases
FROM ads a
JOIN ad_events e ON a.ad_id = e.ad_id
WHERE event_type = 'purchase'
GROUP BY a.ad_type
ORDER BY purchases DESC;
```

Query Result



ad_type	purchases
Stories	589
Carousel	448
Image	408
Video	285

Insight & Recommendation

 Insight: Stories also leads purchases (589) consistent with its engagement dominance. However Carousel has a relatively strong purchase count (448) vs its engagement rank (3rd), suggesting Carousel has a better engagement-to-purchase conversion rate than Image.

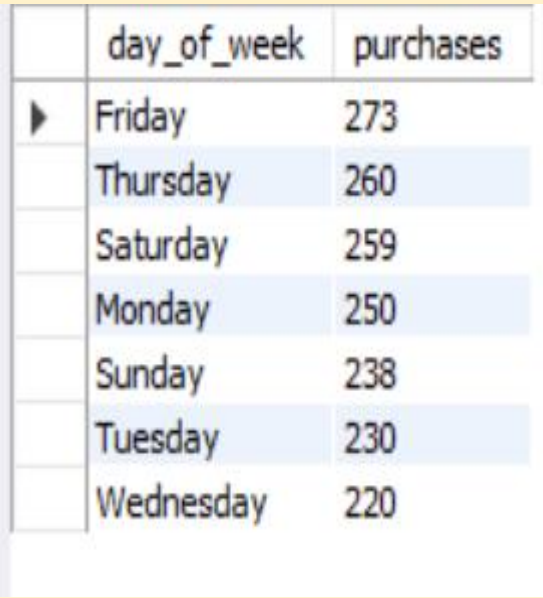
✅**Recommendation:** For conversion campaigns, use Stories first, then Carousel. Carousel's multi-product display naturally supports purchase intent by letting users browse options. Add product pricing and 'Shop Now' buttons to Carousel ads.

Q33. Purchase by Day of Week

SQL Query

```
SELECT day_of_week,
       COUNT(*) AS purchases
FROM ad_events
WHERE event_type = 'purchase'
GROUP BY day_of_week
ORDER BY purchases DESC;
```

Query Result



day_of_week	purchases
Friday	273
Thursday	260
Saturday	259
Monday	250
Sunday	238
Tuesday	230
Wednesday	220

Insight & Recommendation

 InsightFriday leads purchases (273) followed by Thursday (260) and Saturday (259). Tuesday recorded the fewest purchases (233) — a gap of approximately 15% compared to Friday, suggesting lower buying intent mid-week.

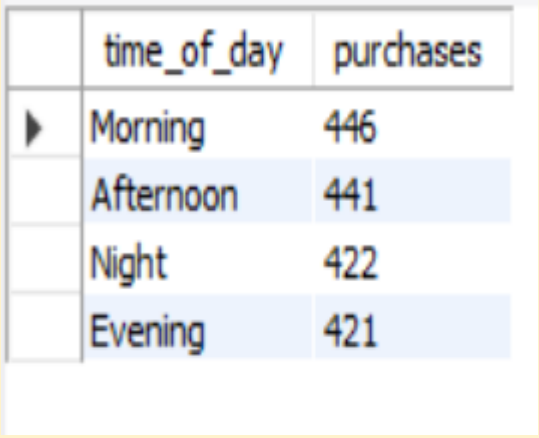
✅**Recommendation:** Schedule purchase-push campaigns (retargeting, discount offers) to peak on Thursday and Friday. Reduce budget on Tuesday or shift Tuesday spend to Thursday/Friday retargeting budgets.

Q34. Purchase by Time of Day

SQL Query

```
SELECT time_of_day,
       COUNT(*) AS purchases
FROM ad_events
WHERE event_type = 'purchase'
GROUP BY time_of_day
ORDER BY purchases DESC;
```

Query Result



time_of_day	purchases
Morning	446
Afternoon	441
Night	422
Evening	421

Insight & Recommendation

 Insight: Morning leads purchases (446) unlike overall engagement where Afternoon led. Afternoon is second (441). Users make purchase decisions in the morning (possibly commuting/browsing before work) and at Afternoon (relaxed home browsing).

✅**Recommendation:** Schedule purchase-intent ads (retargeting + offers) to run during Morning (06:00-12:00) and Night (20:00-24:00). Reduce standard awareness ad spend during Evening and reinvest in Morning purchase campaigns.

Q35. Rank Campaigns by Engagement

SQL Query

```
SELECT c.name,
       COUNT(*) AS engagement,
       RANK() OVER (ORDER BY COUNT(*) DESC) AS campaign_rank
FROM campaigns c
JOIN ads a       ON c.campaign_id = a.campaign_id
JOIN ad_events e ON a.ad_id = e.ad_id
GROUP BY c.name;
```

Query Result

	name	engagement	campaign_rank
▶	Campaign_42_Summer	13717	1
	Campaign_20_Winter	13699	2
	Campaign_17_Launch	13694	3
	Campaign_24_Summer	13690	4
	Campaign_38_Q3	13560	5
	Campaign_33_Summer	12152	6
	Campaign_29_Winter	11880	7
	Campaign_9_Launch	10445	8
	Campaign_13_Winter	10393	9
	Campaign_34_Winter	10374	10

Insight & Recommendation

Insight: **Campaign_42_Summer** has the highest engagement with **13,717**, followed closely by **Campaign_20_Winter** and **Campaign_17_Launch**. The top campaigns have very similar engagement values, showing that several campaigns are performing consistently well.

✓**Recommendation:** Continue investing in these top-performing campaigns and study their strategy, such as audience targeting, creatives, and ad content. Regularly monitor campaign rankings to identify low-performing campaigns and optimize or replace them to improve overall performance.

3. Overall Summary & Strategic Recommendations

3.1 Platform Strategy

- Facebook (63.5% of events) is the primary platform — allocate 60-65% of budget here.
- Instagram (36.5%) serves as secondary channel for brand aesthetics and younger audiences.
- Both platforms have identical funnel conversion efficiency — scale Facebook for volume.

3.2 Ad Format Strategy

- Stories format leads in both engagement (109,816) and purchases (683) — default format.
- Carousel is the best engagement-to-purchase converter after Stories — use for product ads.
- Video has lowest engagement but highest share rate (0.51%) — use for viral/brand campaigns only.
- Image ads are reliable for mid-funnel product awareness.

3.3 Audience Strategy

- Primary audience: Male, 25-34 years — drives 54.5% of engagement and 55.2% of purchases.
- Growth opportunity: Female users are under-converted — needs dedicated creative strategy.
- Geography: USA and UK are established markets; Japan and Mexico have best purchase rates.
- Age 55+: negligible engagement (0.8%) — not worth targeting unless product is specifically for seniors.

3.4 Timing Strategy

- Day-of-week and time-of-day show near-uniform engagement — run always-on campaigns.
- Exception for purchases: Thursday/Friday for day; Morning/Night for time of day.
- Schedule retargeting and discount push ads on Thursday-Friday mornings for best conversion.

3.5 Campaign Performance Benchmarks

Metric	Best Performer	Value	Worst Performer	Value
Total Engagement	Campaign_42_Summer	13,717	Campaign_35_Launch	1,971
Click Rate	Campaign_12_Q3	11.16%	—	—
Purchase Rate	Campaign_27_Q3	0.90%	—	—
Share Rate	Campaign_4_Summer	0.62%	—	—
Engagement/Budget	Campaign_42_Summer	2.02x	—	—

3.6 Conversion Funnel Benchmarks

Funnel Stage	Events	Rate
Impressions	339,812	84.95% of total
Clicks	40,079	10.02% of total | 11.8% of impressions
Likes	12,013	3.00% of total
Comments	4,108	1.03% of total
Purchases	2,031	0.51% of total | 5.1% of clicks
Shares	1,957	0.49% of total

Report generated from: campaigns.csv · ads.csv · users.csv · ad_events.csv | Total records analysed: 400,000 ad events